Simple Network Management Protocol (SNMP):

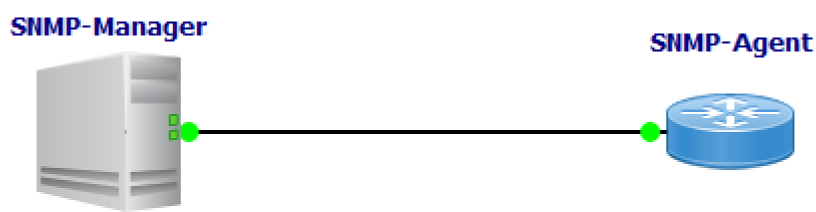
- SNMP is term which is stands for Simple Network Management Protocol.
- SNMP is used to monitor and manage devices on your whole networks.
- It has several uses, from monitoring & generating alerts to device configuration.
- Simple Network Management Protocol (SNMP) is the application layer protocol.
- SNMP is the key protocol used to retrieve information from the network devices.
- SNMP is used to retrieve information from routers, switches & network servers etc.
- SNMP can be configured as Read-Only mode to retrieve only information from devices.
- SNMP Read-Write mode can be used to retrieve or configure the network devices.
- All the SNMP messages are transported via User Datagram Protocol (UDP).
- SNMP agent receives requests on User Datagram Protocol (UDP) port 161.
- SNMP Traps, information to manager over Port User Datagram Protocol UDP 162.

SNMP Manager:

- A software that runs on the device of the Network administrator System.
- A Computer to monitor network, also called Network Management System.

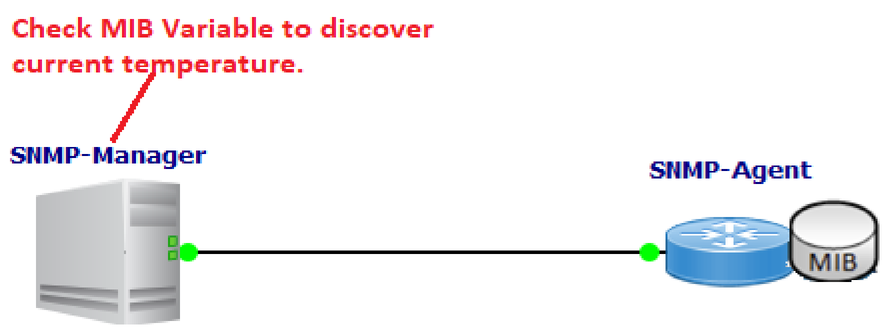
SNMP Agent:

- A software runs on network devices that we want to monitor router, firewall, etc.



Management Information Base (MIB):

- Management Information Base (MIB) is the collection of managed objects.
- MIB contains a set of questions that the SNMP Manager can ask the Agent.
- MIB contains a set of questions that the Agent can understand them.
- MIB is commonly shared between the Agent and the SNMP Manager.



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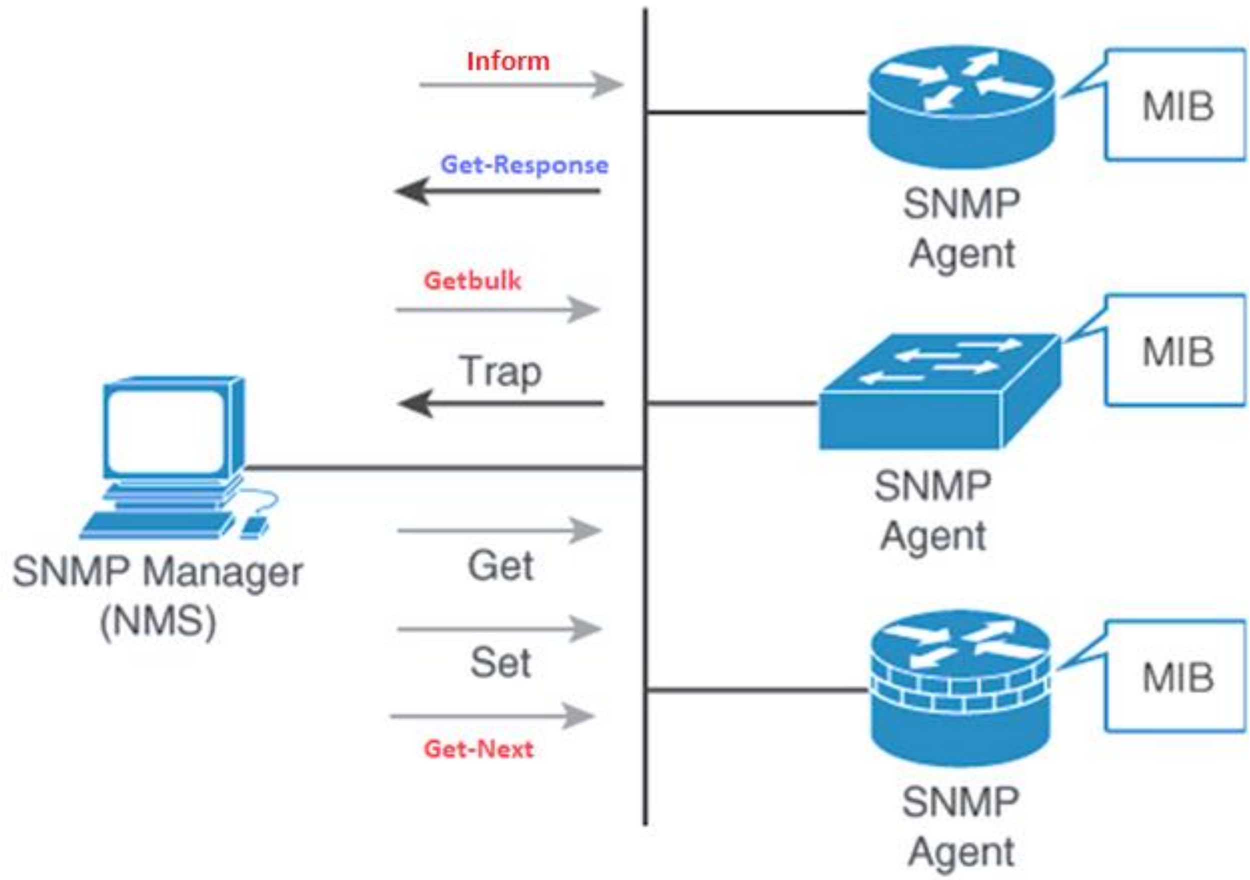
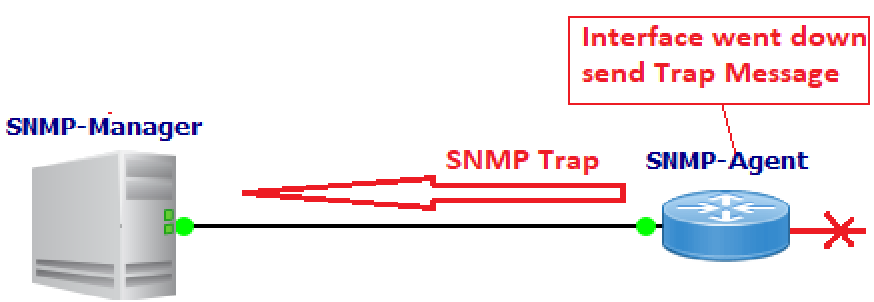
SNMP Messages:

- SNMP Messages are used to communicate between the SNMP Manager and Agents.
- SNMPv1 supports five basic SNMP messages Get, Get-Next, Get-Response, Set & Trap.
- SNMPv2c, two new messages were added Inform message and Getbulk message.
- GET Messages are sent by the SNMP Manager to retrieve info from SNMP Agents.
- SET Messages are used by the SNMP Manager to assign the value to SNMP Agents.
- GET-NEXT retrieves the value of the next object in the Management Information Base.
- GET-RESPONSE Message is used by SNMP Agents to reply to GET & GET-NEXT messages.
- TRAP Messages are initiated from the SNMP Agents to inform the SNMP Manager on event.
- Inform Message, SNMP Manager acknowledge that the message has been received.
- Getbulk operation efficiently retrieve large blocks of data, such as multiple rows in a table.





SNMP Version 1:

- o An SNMP version 1 security is based on community strings.
- o An SNMP community string can be considered as password.

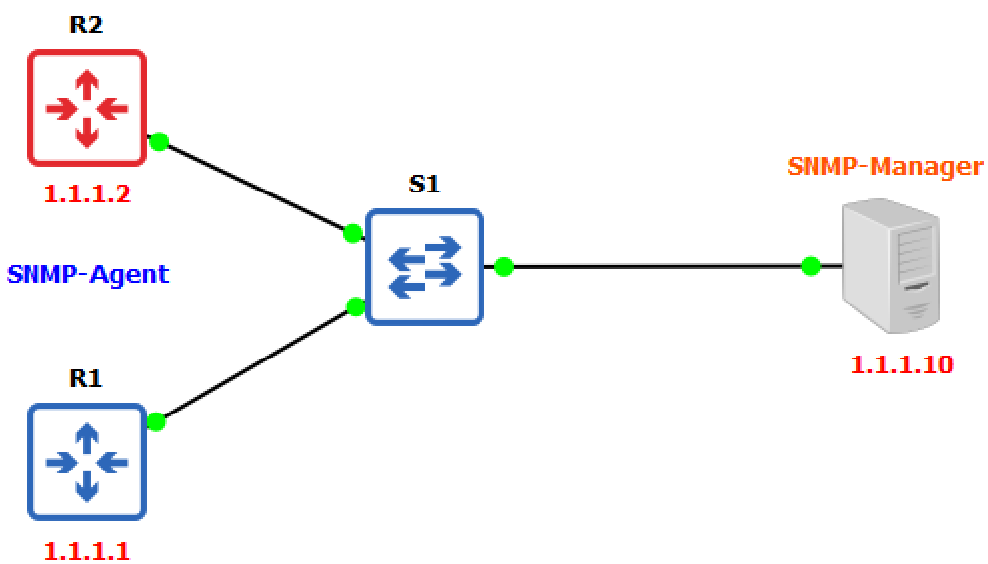
SNMP Version 2c:

- o SNMPv2c is an update SNMPv2 and SNMP Version 1.
- o SNMPv2c uses the community-based security model of SNMPv1.
- o SNMPv2c "c" in SNMPv2c stands for "community".
- o SMMPv2c sends the community strings in clear text.

SNMP Version 3:

- o SNMP Version 3 is the most secure version among other SNMP versions.
- o SNMP V3 provides secure access to devices using authentication & encryption.
- o Authentication security feature makes sure that message is from a valid source.
- o Integrity security feature makes sure that the message has not been tampered.
- o Encryption security feature provides confidentiality by encrypting the contents.
- o SNMP Version 3 will never send the user and the password in the clear text.
- o SNMP Version 3 uses the SHA1 or the MD5 hash-based authentication methods.
- o SNMP Version 3 encryption is done using the methods AES, 3DES and the DES.
- o SNMP offers three security levels: noAuthNoPriv, AuthNoPriv and AuthPriv.
- o Auth stands for Authentication and Priv stand for Privacy in SNMP Version 3.
- o NoAuthNoPriv = no authentication & no encryption of the messages send & receive.
- o AuthNoPriv = authentication but no encryption of the messages send and receive.
- o AuthPriv = authentication necessary and messages are encrypted send and receive.

Lab Time:



Devices Basic Configuration
R1(config)#interface ethernet 0/0 R1(config-if)#ip address 1.1.1.1 255.0.0.0 R1(config-if)#no shutdown
R2(config)#interface ethernet 0/0 R2(config-if)#ip address 1.1.1.2 255.0.0.0 R2(config-if)#no shutdown
S1(config)#interface vlan 1 S1(config-if)#ip address 1.1.1.3 255.0.0.0 S1(config-if)#no shutdown

SNMP Version 2 Configuration
R1(config)# snmp-server community cisco ro
R1(config)# snmp-server community cisco rw
R1(config)# snmp-server location DC
R1(config)# snmp-server contact admin
R1(config)# snmp-server host 1.1.1.10 version 2c test
R1(config)# snmp-server enable traps
R1# show snmp group
R1# show snmp user
R1# show snmp engine ID

SNMP Version 3 No Authentication & Privacy
R1(config)# snmp-server group group1 v3 noauth
R1(config)# snmp-server user user1 group1 v3
R1(config)# snmp-server enable traps
R1(config)# snmp-server host 1.1.1.10 user1

SNMP Version 3 Authentication & No Privacy Configuration
R1(config)# snmp-server group group1 v3 auth
R1(config)# snmp-server user user1 group1 v3 auth md5 authpass
R1(config)# snmp-server enable traps
R1(config)# snmp-server host 1.1.1.10 user1

SNMP Version 3 Authentication & Privacy Configuration
R1(config)# snmp-server group group1 v3 auth
R1(config)# snmp-server user user1 group1 v3 auth md5 authpass
R1(config)# snmp-server enable traps
R1(config)# snmp-server host 1.1.1.10 user1
R1# show snmp group
R1# show snmp user
R1# show snmp

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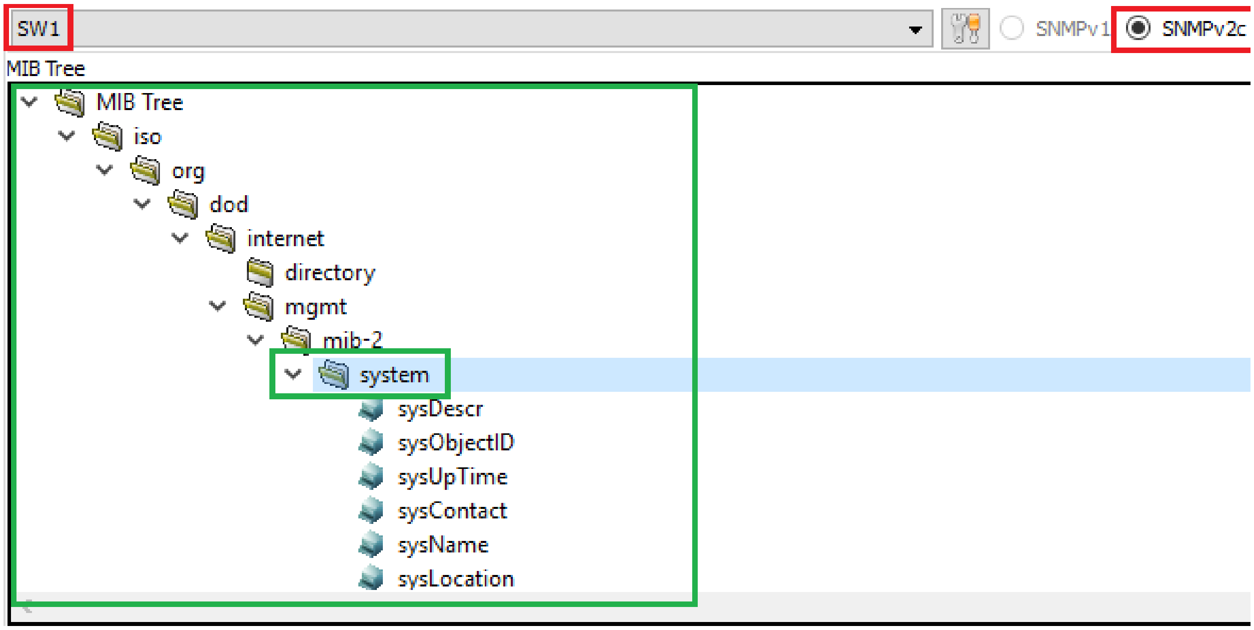
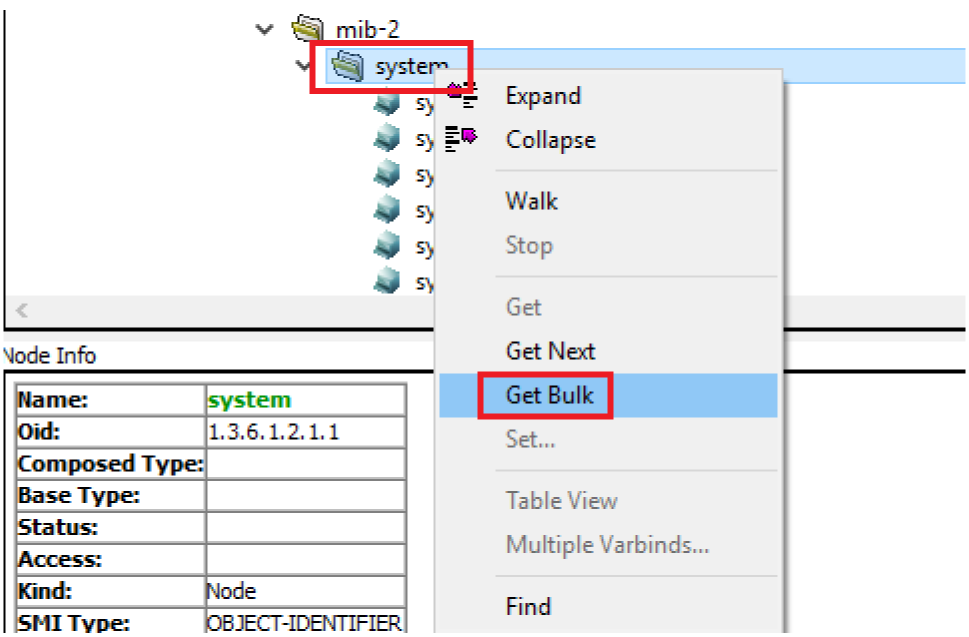
SNMP Messages:

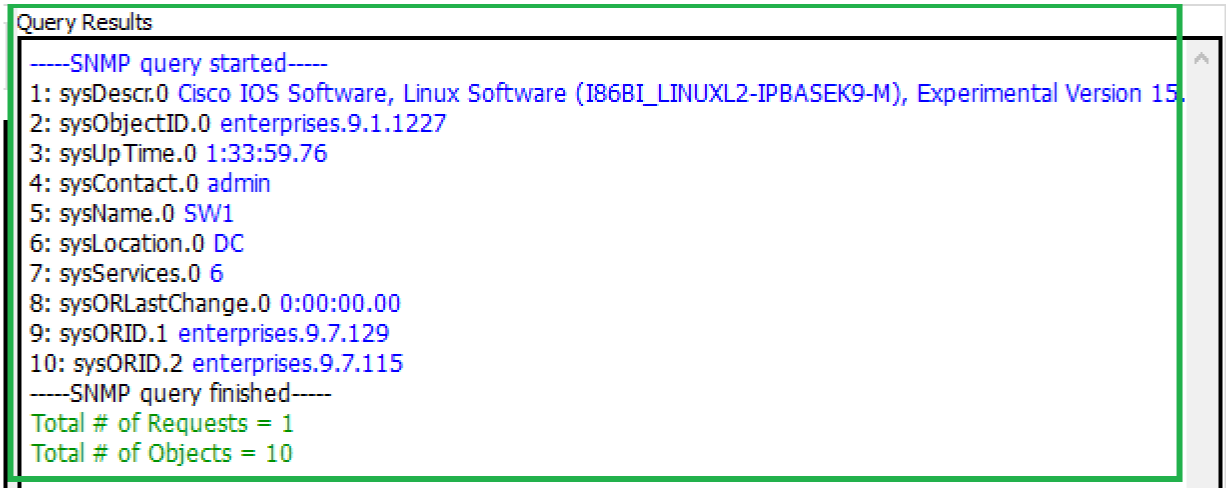
Get, Get-Next, Get-Response, Set and Getbulk.



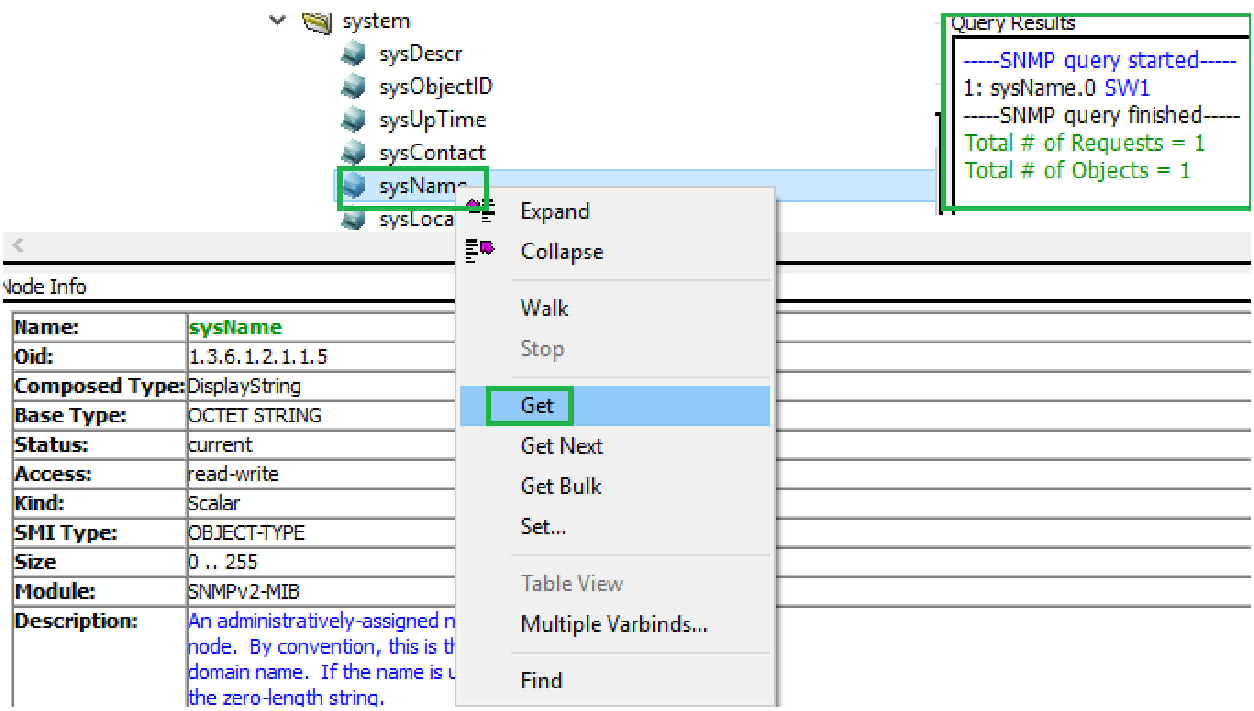
Getbulk operation efficiently retrieve large blocks of data, such as multiple rows in a table.



Getbulk operation efficiently retrieve large blocks of data, System Description, System UP Time, System Contact, and System name, System Location, System Services etc.



GET Messages are sent by the SNMP Manager to retrieve info from SNMP Agents.



SET Messages are used by the SNMP Manager to assign the value to SNMP Agents.